ReMeDi® e-Sanjeevani

Technical Specification Document

Point-of-care diagnostics integrated with the national e-Sanjeevani teleconsultation platform (MoHFW / C-DAC)

Neurosynaptic Communications Pvt. Ltd.

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Document Control

Item	Details
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1. Introduction

1.1 Purpose

This document describes the technical specification of the ReMeDi e-Sanjeevani solution: its architecture, workflow, mobile application, connected medical devices, integration with the e-Sanjeevani platform, security, and validation status. It is intended for review by programme officials and technical evaluators.

1.2 Product Overview

ReMeDi e-Sanjeevani enables a Community Health Officer (CHO) or Doctor using the e-Sanjeevani teleconsultation platform to request point-of-care diagnostic tests for a patient. The request is delivered to the ReMeDi e-Sanjeevani Android application used by the health worker, who performs the tests using ReMeDi medical devices. Results, reports and recordings are returned automatically to e-Sanjeevani and are available to the clinician within the patient's case.

1.3 Key Highlights

- Live integration with e-Sanjeevani; completion of integration confirmed by C-DAC (February 2025).
- 27 diagnostic parameters across physiology, fetal care, blood chemistry and rapid / urine tests.
- Diagnostic request initiated directly from the e-Sanjeevani Clinical Examination section – no separate portal needed by the clinician.
- Results, PDF reports (ECG, spirometry, imaging) and stethoscope audio returned automatically to e-Sanjeevani.
- Android application distributed through the Google Play Store; validated on Android 12, 13 and 14 tablets and smartphones.
- Validation testing completed and documented, with a requirement traceability matrix.

2. System Architecture

The solution has four layers: the e-Sanjeevani platform, ReMeDi Cloud, the ReMeDi e-Sanjeevani Android application at the point of care, and the ReMeDi medical devices.

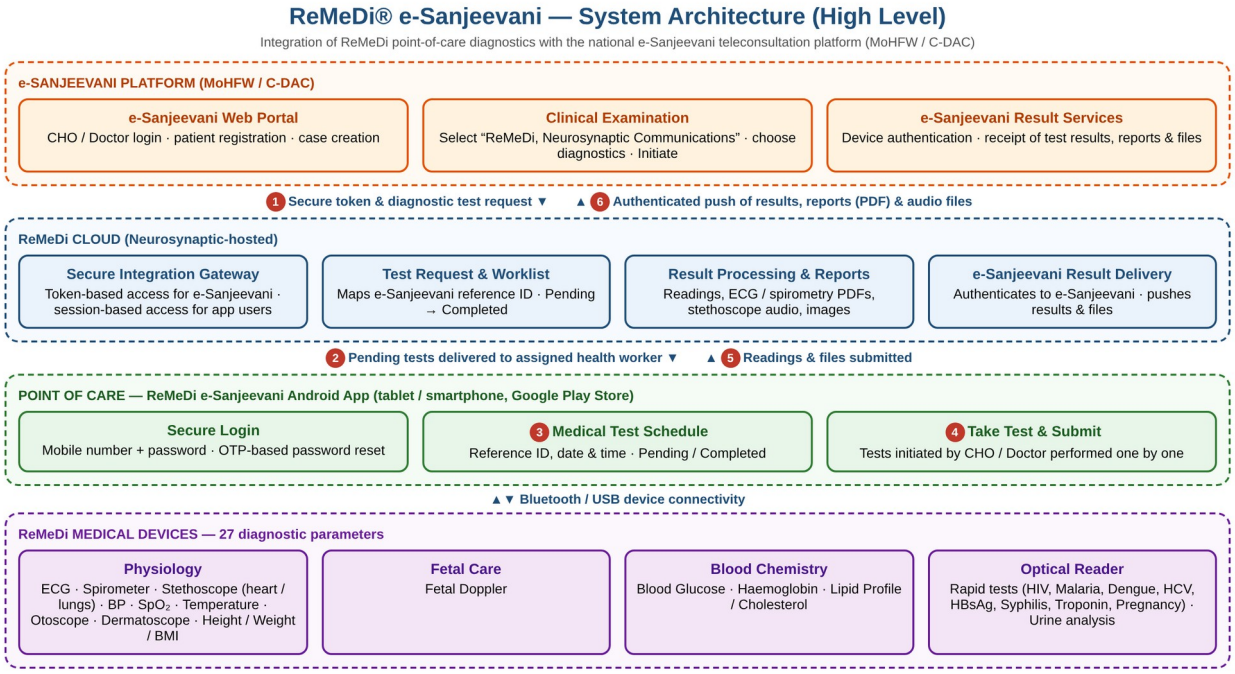


Figure 1 – ReMeDi e-Sanjeevani high-level architecture

2.1 Architecture Components

Layer	Component	Function
e-Sanjeevani (MoHFW / C-DAC)	Web portal & Clinical Examination	CHO / Doctor registers the patient, creates a case and initiates ReMeDi diagnostics.
e-Sanjeevani (MoHFW / C-DAC)	Result services	Authenticates ReMeDi and receives test results, reports and files.
ReMeDi Cloud	Secure integration gateway	Token-based access for e-Sanjeevani; session-based access for app users.
ReMeDi Cloud	Test request & worklist	Records each request against the e-Sanjeevani reference ID and tracks it from Pending to Completed.
ReMeDi Cloud	Result processing & reports	Stores readings and generates / stores ECG and spirometry PDFs, images and audio.
ReMeDi Cloud	Result delivery	Authenticates to e-Sanjeevani and pushes results and files to the patient case.
Point of care	ReMeDi e-Sanjeevani Android app	Health worker login, medical test schedule, guided test capture and submission.
Point of care	ReMeDi medical devices	Capture of physiological and laboratory parameters over Bluetooth / USB.

2.2 End-to-End Data Flow

Step	Description
1	CHO / Doctor initiates diagnostics in e-Sanjeevani; the request (patient details, tests, health worker's phone number and reference ID) is sent securely to ReMeDi Cloud.
2	ReMeDi Cloud adds the request to the assigned health worker's pending list.
3	The health worker sees the request in the Medical Test Schedule (reference ID, date and time).
4	The health worker selects "Take Test" and performs each requested test with the connected ReMeDi devices.
5	Readings and files are submitted to ReMeDi Cloud and the request is marked Completed.
6	ReMeDi Cloud authenticates to e-Sanjeevani and delivers the results, PDF reports and audio files to the patient case.

3. Functional Workflow

3.1 e-Sanjeevani Portal (CHO / Doctor)

- Open the e-Sanjeevani web application and log in as CHO / Doctor.
- Register a new patient (Add Patient) or select a registered patient.
- Expand the patient actions and select Create Case.
- Enter chief complaints, patient history and clinical examination details.
- In Clinical Examination, select "ReMeDi, Neurosynaptic Communications Pvt Ltd".
- Select the required diagnostics and click Initiate to send the request to the health worker.

3.2 ReMeDi e-Sanjeevani App (Health Worker)

- Install the ReMeDi e-Sanjeevani application from the Google Play Store.
- Log in with the registered mobile number and password (OTP-based Forgot Password available).
- View the Medical Test Schedule: patient reference ID with date and time of initiation; use Refresh to update.
- Verify the reference ID and select Take Test; the parameters requested by the CHO / Doctor are displayed.
- Complete each test individually and select Submit to send results back to e-Sanjeevani.
- Completed requests move to the Completed list, where results can be viewed.

4. Mobile Application Specification

Attribute	Specification
Application	ReMeDi e-Sanjeevani (Android)

Version validated	4.1.3 (build 13)
Operating system	Android 12, 13 and 14
Device form factor	Android tablet or smartphone
Validated devices	Samsung Galaxy S21 FE, Galaxy S23, Galaxy Tab A7, Tab A8, Tab A9
Hardware (validated configuration)	8 GB / 16 GB RAM, Snapdragon processor
Network	4G LTE / 5G / Wi-Fi
Device connectivity	Bluetooth / USB to ReMeDi medical devices
Distribution	Google Play Store
User roles	Health worker (Nurse / CHO operator); requests are initiated by CHO / Doctor in e-Sanjeevani
Core functions	Login, OTP-based password reset, pending / completed test lists, guided test capture, result submission, result and report viewing

5. Medical Devices and Diagnostic Parameters

The ReMeDi kit devices fall into four groups:

Group	Devices / Tests
Physiology	Temperature, Pulse Oximeter (SpO_2 / pulse rate), Blood Pressure (sphygmomanometer), ECG, Stethoscope (heart and lungs), Spirometer, Otoscope, Dermatoscope, Height, Weight, BMI, Heart Rate
Fetal Care	Fetal Doppler
Blood Chemistry	Blood Glucose, Haemoglobin, Lipid Profile / Cholesterol
Optical Reader (Rapid & Urine)	Rapid tests: Pregnancy, HIV, Malaria, Dengue, Syphilis, Troponin, Hepatitis C (HCV), Hepatitis B (HBsAg); Urine test: 10 urine parameters

5.1 Parameters Available through e-Sanjeevani

#	Physiology Tests & Scope	Blood & Urine Chemistry Tests
1	Fetal Doppler	Cholesterol
2	Spirometer	Blood Glucose
3	Temperature	Dengue
4	Heart Rate	HCV
5	Otoscope	Hepatitis B
6	Stethoscope – Heart	Malaria
7	Blood Pressure (Sys/Dia)	Hepatitis C
8	Dermascope	Troponin
9	Height	HIV
10	Pregnancy	Urine Checkup
11	Stethoscope – Lungs	Hemoglobin
12	BMI	Lipid Profile
13	ECG	Pulse Oximeter (SpO_2/Pulse Rate)

14	Weight	Syphilis

Detailed device specifications are provided in the enclosed "Technical Specifications – ReMeDi Devices" document.

5.2 Result Formats

Test type	Data returned to e-Sanjeevani
Vitals and numeric tests (BP, SpO_2, temperature, glucose, etc.)	Structured readings with units
ECG, Spirometer	Structured readings and PDF report
Dermatoscope, Otoscope	PDF report with captured images
Stethoscope (heart / lungs)	Audio recording (WAV)

6. Integration Specification

ReMeDi Cloud and e-Sanjeevani communicate through secure, authenticated web service interfaces in both directions. Each request is identified by the e-Sanjeevani reference ID so that results are always associated with the correct patient case.

Interface	Direction	Purpose	Security
Access token	e-Sanjeevani → ReMeDi	Authenticates e-Sanjeevani before requests are sent	Client credentials; time-bound bearer token
Test request	e-Sanjeevani → ReMeDi	Sends patient details, requested tests and the health worker's phone number	Bearer token
Health worker app services	App ↔ ReMeDi	Login, OTP verification, pending / completed lists, result submission, report retrieval	Authenticated user session
Device token	ReMeDi → e-Sanjeevani	Authenticates ReMeDi before results are delivered	Credentials issued by e-Sanjeevani
Result & file delivery	ReMeDi → e-Sanjeevani	Delivers test results, PDF reports and audio files to the case	Device token

Detailed API documentation is available on request.

7. Security and Data Protection

- Role-based access control: only authorised CHOs / Doctors initiate tests and only registered health workers perform them.
- Mutual authentication: separate credentials are used for e-Sanjeevani → ReMeDi and ReMeDi → e-Sanjeevani calls.

- Time-bound access tokens and authenticated user sessions on the mobile application.
- OTP verification on the registered mobile number for password reset.
- Patient results are linked to the e-Sanjeevani reference ID, so results go only to the originating case.
- Encrypted data transmission between all components.
- Periodic data backup and restore, verified during validation testing.

8. Non-Functional Specification

Attribute	Specification
Performance	Application response time verified under normal and high load (RTM TC-701)
Scalability	Cloud-hosted backend designed to scale with growing numbers of users, facilities and devices
Availability	Target 99.9% uptime with failover mechanisms
Data management	Periodic backup and restore of patient records (RTM TC-901)
Usability	Guided, step-by-step test capture designed for frontline health workers
Connectivity	Operates over 4G LTE, 5G or Wi-Fi

9. Validation and Testing

Item	Details
Integration status	Integration with e-Sanjeevani completed; confirmed by C-DAC (February 2025)
Test report	ReMeDi-eSanjeevani Test Report, prepared for COEP (College of Engineering Pune), 07 March 2025
Testing period	3 June 2024 to 26 August 2024
Test cases	86 test cases executed; testing reported 100% complete
Test types	Functional, integration, security, performance and UI/UX testing – all passed
Diagnostic parameters tested	All 27 parameters listed in Section 5.1 – passed
Traceability	Requirement Traceability Matrix maps authentication, device connectivity, security (RBAC), performance and data management requirements to passed test cases

9.1 Requirement Traceability Summary

Req. ID	Requirement	Test Case	Status
FR-001	User login	TC-101	Pass
FR-002	Forgot password recovery	TC-102	Pass
FR-005	Medical device integration (BP, ECG,	TC-401	Pass

	glucose)		
FR-007	Role-based access control	TC-601	Pass
FR-008	Application response time	TC-701	Pass
FR-010	Data storage & backup	TC-901	Pass

10. Deployment and Operating Requirements

- Android tablet or smartphone (Android 12 or above) with the ReMeDi e-Sanjeevani app installed from the Google Play Store.
- ReMeDi medical device kit for the tests to be offered at the facility.
- Internet connectivity (4G / 5G / Wi-Fi) at the point of care.
- Health worker accounts registered against their mobile numbers.
- Active e-Sanjeevani CHO / Doctor accounts to initiate diagnostics.
- No additional server infrastructure is required at the facility; the backend is hosted on ReMeDi Cloud.

11. Enclosures

#	Document
A	Technical Specifications – ReMeDi Devices
B	ReMeDi e-Sanjeevani Workflow and Block Diagram
C	C-DAC email confirming completion of integration (07 February 2025)
D	ReMeDi-eSanjeevani Test Report (07 March 2025)
E	Requirement Traceability Matrix – e-Sanjeevani
F	ReMeDi e-Sanjeevani operations video (link)